LAB EXPERIMENTS

Q15) Write a C program that can perform a letter frequency attack on an additive cipher without human intervention. Your software should produce possible plaintexts in rough order of likelihood. It would be good if your user interface allowed the user to specify "give me the top 10 possible plaintexts."

Aim:

To write a C program to decrypt a ciphertext encrypted using the **Playfair Cipher** by constructing a **5 × 5 key matrix** and applying the Playfair decryption rules.

Algorithm:

1. Start the program.
2. Read the Playfair cipher text from the user.
3. Enter the keyword and construct the 5 × 5 Playfair matrix by removing duplicate letters and treating **I/J** as the same character.
4. Divide the ciphertext into pairs of letters (digraphs).
5. Locate the positions of both letters of each pair in the key matrix.
6. Apply the Playfair decryption rules:
    - If both letters are in the same row, replace each with the letter to its left.
    - If both letters are in the same column, replace each with the letter above it.
    - Otherwise, replace each letter with the letter in the same row but in the other letter's column.
7. Repeat the process for all digraphs.
8. Combine all decrypted pairs to obtain the plaintext.
9. Display the decrypted message.
10. Stop the program.

Result:

Thus, the C program to decrypt the given **Playfair Cipher** was successfully executed using a **5 × 5 key matrix**, and the corresponding plaintext message was obtained by applying the Playfair decryption rules.

Code:

C:\Users\SAIL\Desktop\lab15.cpp - [Executing] - Dev-C++ 5.11

```c
#include <stdio.h>
#include <string.h>
#include <ctype.h>
char matrix[5][5];
void findPosition(char ch, int *row, int *col)
{
    if(ch == 'J')
        ch = 'I';
    for(int i = 0; i < 5; i++)
    {
        for(int j = 0; j < 5; j++)
        {
            if(matrix[i][j] == ch)
            {
                *row = i;
                *col = j;
                return;
            }
        }
    }
}
int main()
{
    char key[] = "PLAYFAIR";
    char alphabet[] = "ABCDEFGHIKLMNOPQRSTUVWXYZ";
    int used[26] = {0};
    int k = 0;
    for(int i = 0; key[i] != '\0'; i++)
    {
        char ch = toupper(key[i]);
        if(ch == 'J') ch = 'I';
        if(!used[ch - 'A'])
        {
            matrix[k / 5][k % 5] = ch;
            used[ch - 'A'] = 1;
            k++;
        }
    }
    for(int i = 0; alphabet[i] != '\0'; i++)
    {
        char ch = alphabet[i];
        if(!used[ch - 'A'])
        {
            matrix[k / 5][k % 5] = ch;
```

Line: 84 Col: 6 Sel: 0 Lines: 86 Length: 2032 Insert Done parsing in 0.016 seconds

C:\Users\SAIL\Desktop\lab15.cpp - [Executing] - Dev-C++ 5.11

```c
        {
            matrix[k / 5][k % 5] = ch;
            used[ch - 'A'] = 1;
            k++;
        }
    }
    char cipher[500];
    printf("Enter Cipher Text:\n");
    fgets(cipher, sizeof(cipher), stdin);
    printf("\nDecrypted Text:\n");
    for(int i = 0; cipher[i] != '\0'; i += 2)
    {
        if(cipher[i] == ' ')
        {
            printf(" ");
            i--;
            continue;
        }
        char a = toupper(cipher[i]);
        char b = toupper(cipher[i + 1]);
        int r1, c1, r2, c2;
        findPosition(a, &r1, &c1);
        findPosition(b, &r2, &c2);
        if(r1 == r2)
        {
            printf("%c%c",
                matrix[r1][(c1 + 4) % 5],
                matrix[r2][(c2 + 4) % 5]);
        }
        else if(c1 == c2)
        {
            printf("%c%c",
                matrix[(r1 + 4) % 5][c1],
                matrix[(r2 + 4) % 5][c2]);
        }
        else
        {
            printf("%c%c",
                matrix[r1][c2],
                matrix[r2][c1]);
        }
    }
    return 0;
}
```

Line: 84 Col: 6 Sel: 0 Lines: 86 Length: 2032 Insert Done parsing in 0.016 seconds

Output:

C:\Users\SAIL\Desktop\lab15.exe

```
Enter Cipher Text:
KXJEY UREBE ZWEHE WRYTU HEYFSKREHE GOYFI WTTTU OLKSY CAJPOBOTEI ZONTX BYBNT GONEY CUZWRGDSON SXBOU YWRHE BAAHY USEDQ

Decrypted Text:
CSPIPKVIHIHIXVMGMGVBFSZNGMAYKCIGGM RGAYDPZQSSZNGVCKXKBYPUQRNSIP VTTSZSCAIQSTRGEIPKIXXVLRCTNT KSRQVNAXBGMGAWWBAKXNMITB
--------------------------------
Process exited after 3.043 seconds with return value 0
Press any key to continue . . .
```